Question

Write a menu-driven C program to perform the following operations using separate functions:

- Reverse each word in a sentence without changing the word order. Input: What is special?, Output: tahw si ?laiceps
- Print the initials of a name. E.g. Input: Alexander Graham bell, Output: A.G.

1. Problem Understanding

The program is menu-driven and performs two string operations using separate functions. It can reverse every word of a sentence while keeping the word order unchanged, and it can print the initials of a name.

2. Standard Test Cases

S.No.	All Inputs	Expected Output
1	Choice = 1 Sentence: What is special?	tahw si ?laiceps
2	Choice = 2 Name: Alexander Graham bell	A.G.
3	Choice = 1 Sentence: Hello World	olleH dlroW

3. Algorithm

1. Start.
2. Display the menu.
3. Read the user choice.
4. If choice is 1, read a sentence and call reverseWords().
5. In reverseWords(), reverse the characters of each word separately without changing the order of words.
6. If choice is 2, read a name and call printInitials().
7. In printInitials(), print the first character of each name part followed by a dot. Do not print the last name initial, as shown in the given example.
8. If choice is 3, exit the program.
9. For any other choice, display an invalid choice message.
10. Stop.

4. C Program

```
#include <stdio.h>
#include <string.h>

void reverseWords(char str[])
{
  int i = 0, start, end;
  char temp;

  while (str[i] != '\0')
  {
```

```
    while (str[i] == ' ')
      i++;

    start = i;
    while (str[i] != ' ' && str[i] != '\0')
      i++;
    end = i - 1;

    while (start < end)
    {
      temp = str[start];
      str[start] = str[end];
      str[end] = temp;
      start++;
      end--;
    }
  }
}

void printInitials(char name[])
{
  int i, words = 0, current = 0;

  for (i = 0; name[i] != '\0'; i++)
    if ((i == 0 && name[i] != ' ') || (name[i] != ' ' && name[i - 1] == ' '))
      words++;

  for (i = 0; name[i] != '\0'; i++)
  {
    if ((i == 0 && name[i] != ' ') || (name[i] != ' ' && name[i - 1] == ' '))
    {
      current++;
      if (current < words)
        printf("%c.", name[i]);
    }
  }
  printf("\n");
}

int main()
{
  int choice;
```

```
    char str[200];

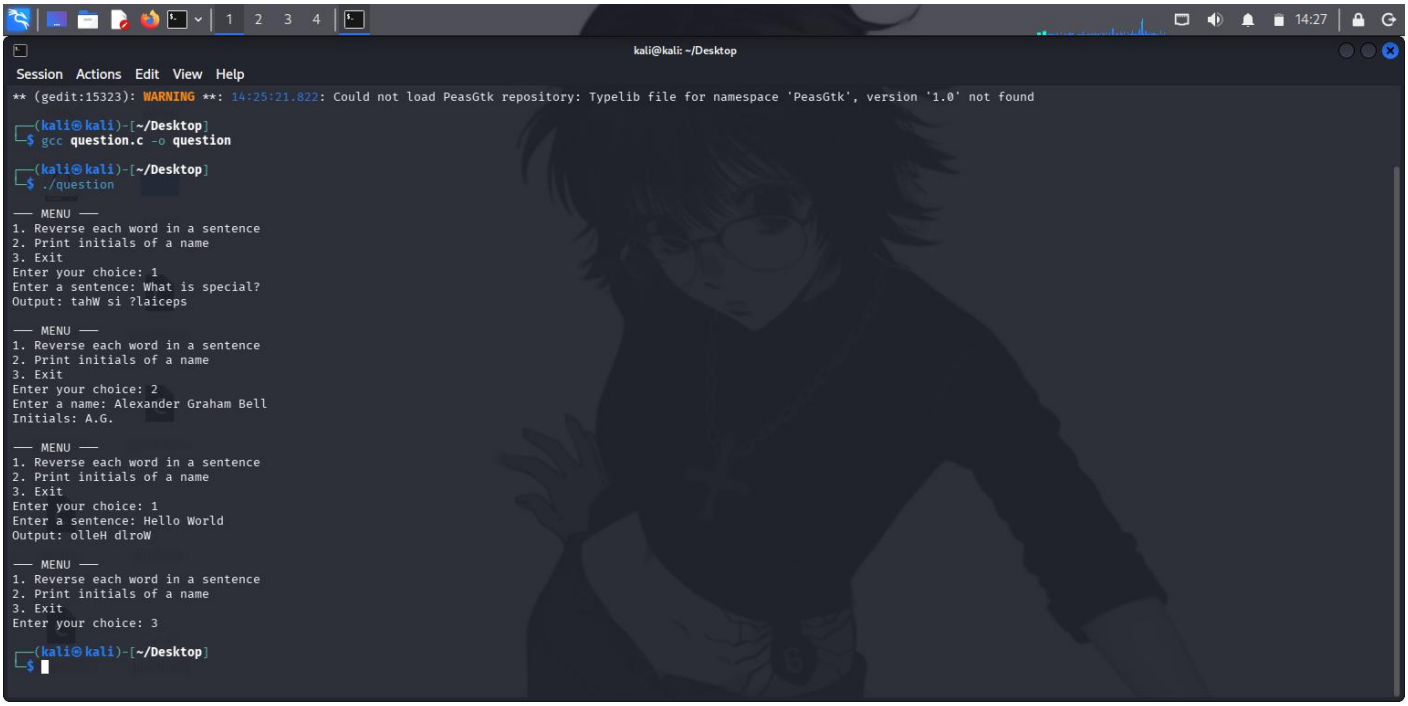
    do
    {
        printf("\n--- MENU ---\n");
        printf("1. Reverse each word in a sentence\n");
        printf("2. Print initials of a name\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();

        if (choice == 1)
        {
            printf("Enter a sentence: ");
            fgets(str, sizeof(str), stdin);
            str[strcspn(str, "\n")] = '\0';
            reverseWords(str);
            printf("Output: %s\n", str);
        }
        else if (choice == 2)
        {
            printf("Enter a name: ");
            fgets(str, sizeof(str), stdin);
            str[strcspn(str, "\n")] = '\0';
            printf("Initials: ");
            printInitials(str);
        }
        else if (choice != 3)
            printf("Invalid choice.\n");

    } while (choice != 3);

    return 0;
}
```

5. Compilation and Execution Evidence



6. Verification of All Test Cases

Test	Input Summary	Actual Output	Result
1	What is special?	tahw si ?laiceps	Pass
2	Alexander Graham bell	A.G.	Pass
3	Hello World	olleH dlroW	Pass